



A lightweight multi-view detection and counting method for real-time cherry tomato yield estimation using greenhouse inspection robots

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ABSTRACT

Crop yield prediction is one of the key measures for dynamically adjusting harvest schedules and minimizing production losses. To address the production requirements for multi-object real-time detection and accurate counting of tomatoes in complex greenhouse environments, this study proposes a lightweight “detection–tracking–counting–compensation” cascaded network framework, optimized for deployment on inspection robots. First, a sparse convolution module (G-FConv) and a lightweight multi-scale feature aggregation module (L-MFA) were designed, and a lightweight feature extraction network was constructed to reduce computational load and complexity. Second, a lightweight asymmetric decoupled head incorporating the SimAM attention mechanism was developed to enhance critical feature extraction while further reducing parameter count. Third, a multi-view cooperative detection and compensation method was proposed to optimize multi-object tracking performance in dense and occluded scenes. Finally, TensorRT-based inference acceleration was applied to improve the robot’s real-time processing capability. The proposed model achieves high detection (90.8%) and counting (95.2%) accuracy with a compact size of 3.98 MB and 5.3 GFLOPs, while the proposed MVC method reduces counting errors by 95.6%. This study provides an effective technical solution for precise tomato yield prediction in greenhouses, offering significant practical value for optimizing harvest schedules and reducing production losses.

1. Introduction

Tomato is a globally significant economic crop, with an annual production exceeding 180 million tons cultivated over more than 5 million hectares. Its strategic importance and substantial economic value are well established. The expanding global market, projected to reach US\$261.2 billion by 2030 [1], reflects strong and growing consumer demand. However, current tomato yields frequently fail to align with actual market requirements, resulting in regional oversupply. This imbalance disrupts supply chain stability and undermines brand credibility, leading to considerable economic losses. Accurate yield estimation is therefore critical for maintaining efficient supply chain operations and promoting sustainable agricultural practices. Object detection and tracking are critical techniques to enhance precision

farming and pave the way for robotization in the agricultural sector, as they provide accurate results and insights for crop management while optimizing resource allocation [2].

Traditional object detection methods primarily rely on hand-crafted features and classical image processing. These approaches can be categorized into three types: Color-space-based segmentation (e.g., HSV or Lab thresholding) extracts target regions but is highly sensitive to lighting conditions and color variations [3]. Edge detection and morphological operations (e.g., Canny and Sobel filters combined with erosion/dilation) facilitate contour extraction but are prone to noise and perform poorly with occluded or clustered objects [4]. Feature descriptor-based classifiers (e.g., HOG + SVM) use Histograms of Oriented Gradients (HOG) for texture representation and classification [5]; however, these methods suffer from limited feature expressiveness, poor

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generalization across diverse scenes, and often require crop-specific parameter tuning. While traditional methods offer advantages in simple scenarios—such as lower computational costs and reduced latency—they face critical limitations in complex, unstructured agricultural settings. These challenges include extensive reliance on expert-driven parameter adjustments, difficulties in segmenting densely occluded or overlapping targets, and limited adaptability to fluctuating lighting conditions. Collectively, these constraints significantly hinder the robustness and scalability of traditional detection systems in real-world agricultural applications.

Traditional counting methods predominantly rely on hand-crafted features, such as Histogram of Oriented Gradients (HOG) and color histograms for pedestrian tracking [6]. However, these methods exhibit limited generalization in dynamic environments and are highly susceptible to noise interference during data association. Given the inherent limitations of hand-crafted features, Multi-Object Tracking (MOT) methods have become prevalent, following a "detection-feature extraction-data association" framework. The performance of such methods is heavily dependent on the accuracy of object detection, the discriminative power of extracted features, and the robustness of the association strategy. In complex scenes, challenges such as object occlusion, deformation, appearance variation, similar-object interference leading to identity switches (ID switches), and real-time processing requirements significantly exacerbate tracking difficulties, posing serious challenges for traditional counting methods [7]. The OC-SORT algorithm proposed by Cao et al. [8] introduces a stable tracking framework by integrating Kalman filtering, the Hungarian algorithm, and multiple association recovery strategies. Building on this, Deep OC-SORT [9] incorporates stability-enhancing mechanisms to improve adaptability in complex scenarios. Nevertheless, substantial challenges remain in handling occlusion, object deformation, appearance changes, and interference from similar targets, which continue to limit the performance gains of current approaches and present bottlenecks to further advancement.

The rapid progression of deep learning technologies is driving a transformative shift towards automation and intelligence in tomato cultivation [10], with researchers increasingly integrating these methodologies for crop yield estimation. For instance, while Fast R-CNN [11] offers high detection accuracy, its substantial computational resource demands and deployment complexity present practical challenges. Similarly, the SSD model [12], reduces the computational load, but its intricate parameter tuning and sensitivity to small objects limit its efficacy for yield estimation, making it more suitable for small-target applications like pest and disease monitoring [13]. Tenorio and Caarls [14] developed an automated tomato detection model capable of classifying fruit maturity levels. Xu et al. [15] proposed an edge-friendly framework that enhances inference efficiency while maintaining accuracy. Sapkota et al. [16] developed an instance segmentation algorithm for robotic tasks. Gao et al. [17] introduced an unstructured detection framework that enables seamless integration into harvesting robots. Paul et al. [18] presented a complete solution for agricultural robots that integrates detection, localization, tracking, and application, thereby advancing the development of automated capsicum harvesting technology. These approaches provide feasible solutions for tomato detection, counting, and algorithm deployment.

Tomato detection and counting remain challenging tasks, especially in complex greenhouse environments. Specifically, truss tomatoes in greenhouse settings often exhibit severe fruit overlap and mutual occlusion, leading to blurred object boundaries and difficulty in distinguishing individual fruits. Meanwhile, the image acquisition process introduces significant motion blur due to robot movement or fruit cluster swaying, and the pose variations of tomatoes further increase detection difficulty. These factors collectively make it difficult for traditional detection and tracking methods to meet practical requirements in terms of accuracy and robustness. Therefore, there is an urgent need for a lightweight detection framework that balances

accuracy, speed, and computational efficiency, especially one that can effectively address overlapping occlusion, motion blur, and pose variation. However, the large parameter size and high computational demands of deep learning models hinder their deployment on resource-constrained edge devices, and existing edge intelligence techniques still face challenges in interpretability, security, and energy efficiency [19]. At present, counting algorithms specifically designed for such highly challenging scenarios are still lacking.

To address the above challenges, this study proposes a lightweight cascaded neural network for tomato detection and counting. By optimizing the network architecture, integrating multi-object real-time tracking algorithms, and developing a multi-view compensation method, the proposed framework achieves real-time, efficient, and accurate detection and counting of tomatoes in complex greenhouse environments. This offers a promising solution to the labor shortages and inaccurate yield estimation currently faced by the tomato cultivation industry.

2. Data collection and processing

From November 2024 to March 2025, image acquisition of cherry tomatoes was conducted at the Kaisheng Haofeng (Laixi) Smart Agriculture Industrial Park in Qingdao, Shandong Province, China. The tomato cultivation base adopts a modern planting model with standardized plant layouts. Each row consists of two lines of tomato plants growing in opposite directions, and the growth conditions are largely consistent. However, occlusion and overlap among fruits are common. The image acquisition device used was the Intel RealSense Depth Camera D455, which was mounted on a mobile inspection robot. The distance between the camera and the cherry tomato plants ranged from 400 mm to 600 mm. Videos were recorded at a resolution of 1280×720 with a frame rate of 30 FPS. The collected videos encompass cherry tomatoes captured across various planting zones during both morning and afternoon time periods, viewing angles, and environmental conditions, with all weather conditions being sunny. In the sample collection scenarios, normal lighting accounts for approximately 65%, low lighting for about 25%, and strong lighting for about 10%. The sample proportions across different time periods are generally balanced. An overview of the typical greenhouse cultivation scene is shown in Fig. 1.

In this study, the collected video data were divided and processed in two ways. A portion of the data was directly used as the validation set for evaluating the model's detection performance. The remaining video data were processed by the keyframe extraction at 60 FPS intervals. A total of 1792 images were extracted and annotated using the Makesense AI platform [20]. Based on the color and size characteristics of cherry tomatoes, the detected targets are divided into two categories: ripe tomatoes and unripe tomatoes, marked as "rt" and "ut" respectively. The dataset comprises 17,940 "ut" instances and 33,433 "rt" instances. For occluded fruits, annotators inferred the complete boundary based on visible contours and contextual information, drawing the bounding box within the occluded boundary to ensure object completeness under occlusion. As illustrated in Fig. 2, the full contours of occluded tomatoes were annotated to improve the accuracy of object representation. All annotation results were standardized and saved in .txt format, and the dataset was divided into training, validation, and testing subsets in a 7:2:1 ratio.

3. Material and methods

3.1. Development of a multi-task cascaded neural network

To meet the real-time and efficient detection requirements of greenhouse inspection robots, this study proposes a lightweight multi-task cascaded neural network capable of performing real-time multi-object detection and multi-region tracking-based counting. The overall

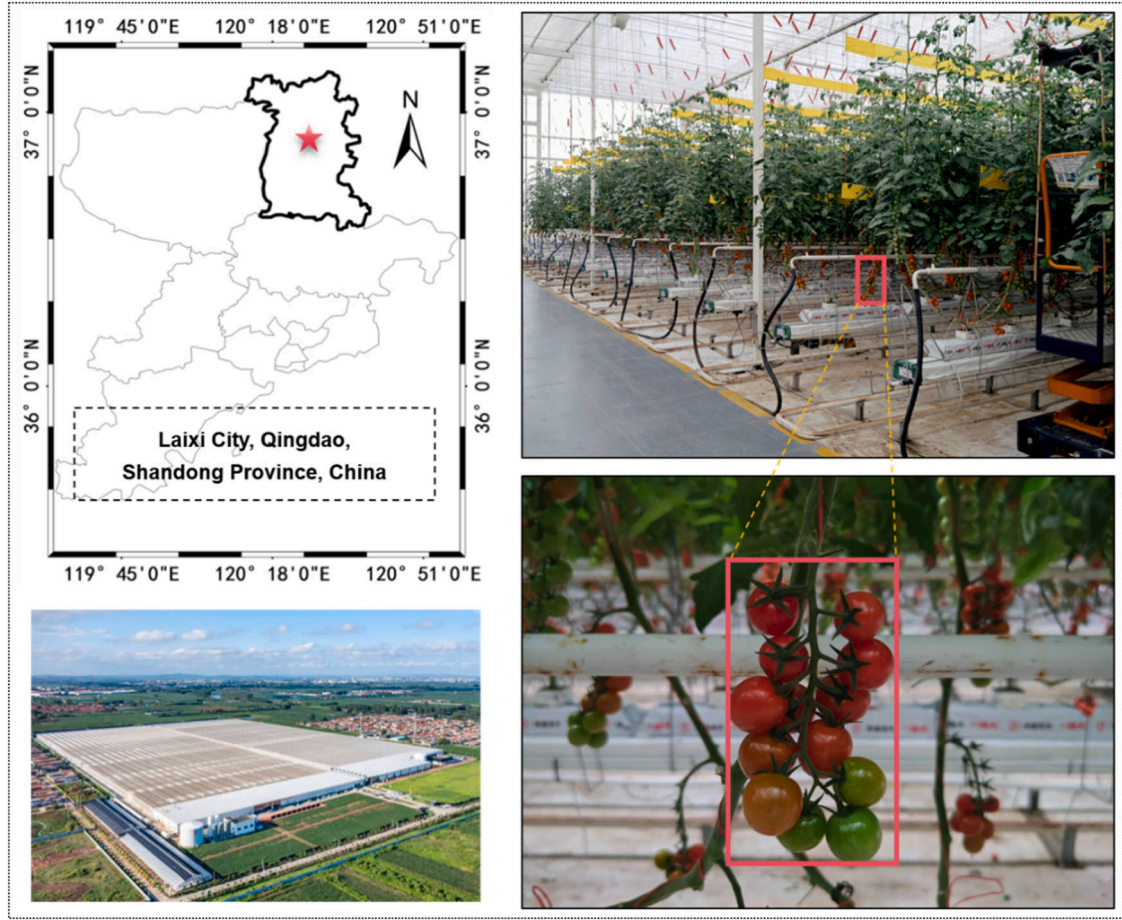


Fig. 1. Typical greenhouse cultivation scene and geographic location.



Fig. 2. Image acquisition platform and data annotation.

architecture, illustrated in Fig. 3, adopts a cascaded "detection–tracking–counting–compensation" framework. Through modular design and optimization, the network achieves a synergistic optimization between detection accuracy and computational efficiency, thereby enhancing performance in complex greenhouse environments.

In operation, the network first integrates a lightweight detection module, which is responsible for identifying the number and type of cherry tomatoes and outputting corresponding confidence scores and class labels. This information is then passed to the Deep OC-SORT tracking network, which assigns a unique ID to each detected object and performs continuous tracking across frames. To address challenges like occlusion and duplicate counting in agricultural scenarios, a multi-view compensation (MVC) network is introduced based on the principles

of 3D spatial projection. By conducting independent detection and integrated analysis across different spatial regions, MVC effectively mitigates counting errors caused by occlusion and overlapping, thereby improving the reliability and accuracy of detection and counting in complex greenhouse environments. The lightweight detection module serves as the front-end, the tracking and multi-view compensation module performs counting, and the edge deployment optimization ensures real-time performance

3.2. Lightweight multi-object real-time detection network

3.2.1. Lightweight feature extraction backbone network

The input image first passes through a GhostConv module for lightweight convolution, generating redundant feature maps via cheap linear transformations to reduce computational cost. It then enters L-MFA, to extract and fuse multi-scale contextual information. Next, the G-FConv module further compresses parameters and computational cost using sparse convolution and linear expansion. This G-FConv is stacked twice to enhance feature abstraction. Finally, the Spatial Pyramid Pooling Fast module, denoted as SPPF, performs multi-scale pooling on the feature maps, aggregating information from different receptive fields and outputting multi-level feature maps such as P3, P4 and P5 for subsequent detection tasks. Among these components, the G-FConv module is specifically designed to reduce computational overhead and memory access, and its architecture is detailed as follows. G-FConv is a sparse convolution module derived from the lightweight GhostNet [21], designed to reduce computational overhead and memory access. The architecture of the proposed module is illustrated in Fig. 4. The core idea is to generate redundant features through linear transformations, while integrating PConv [22] to perform convolution on a subset of the input

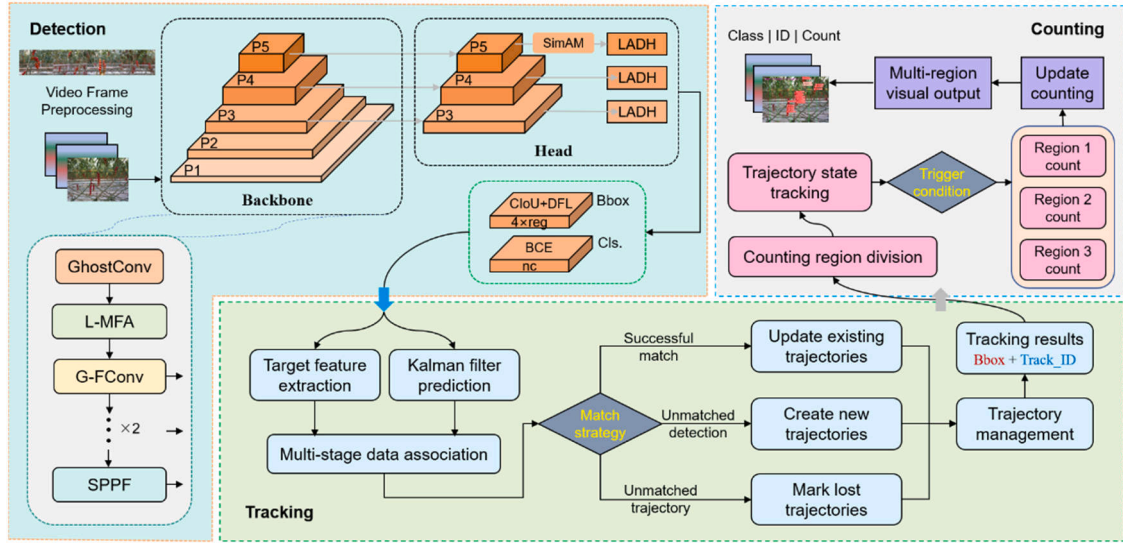


Fig. 3. Architecture of the proposed cascaded network framework for real-time cherry tomato detection and multi-region counting.

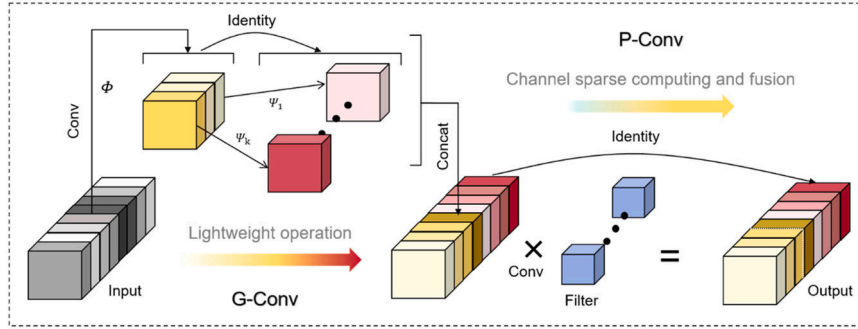


Fig. 4. Network structural design of G-FConv combining PConv-based sparse computing and Ghost-style linear expansion.

channels, thereby reducing computational cost. Furthermore, a co-design strategy combining sparse convolution and linear feature expansion is adopted to enhance feature extraction efficiency. Specifically, in the G-FConv module, the PConv part uses a 3×3 kernel, and its output channels are set to $1/4$ of the input channel. The Ghost linear transformation part uses 1×1 convolutions to generate redundant features with an expansion ratio of 2. The input and output channel numbers of the entire G-FConv module are kept identical to facilitate residual connections. This design allows the network backbone to maintain high accuracy in visual tasks while achieving lightweight characteristics. The computational formulation of its feature extraction is as follows:

$$Y_{Combine} = f_F \cdot f_G \cdot f_F \cdot f_G(X) \quad (1)$$

where, f_G denotes the feature extraction output from the GhostConv module. f_F denotes the feature extraction output from the PConv module. $Y_{Combine}$ denotes the connection strategy of the feature extraction module.

To further reduce model computational complexity and enhance real-time processing performance, this study proposes a lightweight multi-scale feature aggregation module (L-MFA). As illustrated in Fig. 5, the module is primarily composed of Conv and CG Blocks. By concatenating the outputs of different CG Blocks with the original feature maps, the module achieves more effective multi-scale information aggregation. In addition, the convolutional operations within the module efficiently compress the feature maps, substantially reducing computational cost. The L-MFA module adopts the classical Cross-Stage Partial (CSP)

structure characterized by a “split–process–integrate” design, ensuring efficient gradient flow, while incorporating the context-guided (CG) Block to further strengthen feature fusion. The L-MFA module uses a Split operation to directly pass a portion of the features across layers, while the other portion is processed through multiple CG block for deep feature extraction. Finally, the multi-level information is fused via concatenation and a 1×1 convolution. Given an input feature map of size $h \times w \times c_{in}$, it first passes through a 1×1 convolution (stride 1, padding 0) that compresses the number of channels to c_{out} , resulting in an output size of $h \times w \times c_{out}$. Then a Split operation is performed, which evenly divides the feature map into two branches along the channel dimension, each having $0.5 \times c_{out}$ channels, i.e., $h \times w \times 0.5c_{out}$. The left branch remains unchanged and is directly kept for later concatenation. The right branch goes into a sub-network consisting of n CG bBlock modules stacked in series. Each CG Block has an optional shortcut connection and maintains the same spatial size and channel count, thus its output is still $h \times w \times 0.5c_{out}$. Finally, a Concat operation concatenates the original left branch feature map and the n outputs from the right branch along the channel dimension. This design not only inherits the advantages of the CSP architecture but also leverages the superior capability of CGNet [23] in extracting contextual information. To enhance multi-scale contextual information without significantly increasing computation, we introduce the CG Block, which captures local, surrounding, and global features across multiple receptive fields. The CG Block, a key component of the CGNet model, employs a four-dimensional context fusion mechanism to capture and integrate local features, surrounding context, and global context. In doing so, it

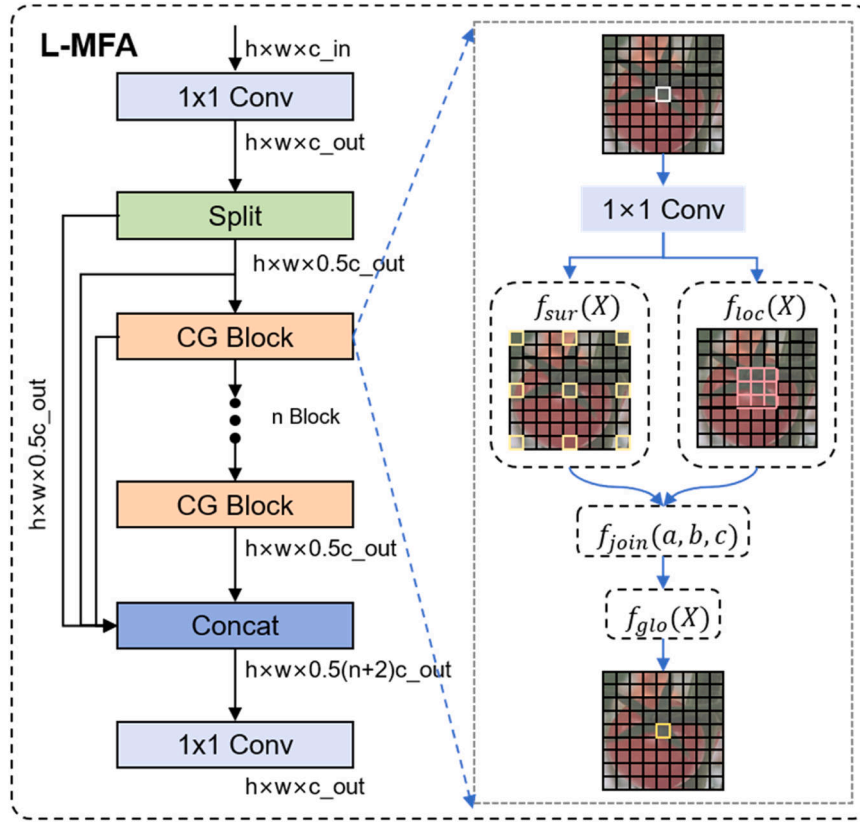


Fig. 5. Network structural composition of the lightweight L-MFA module for efficient multi-scale information fusion.

maintains model accuracy while effectively reducing computational complexity. In the L-MFA module, the CG Block consists of three feature extractors. The local feature extractor applies a 3×3 depthwise convolution (stride 1, padding 1). The surrounding context extractor uses a 3×3 depthwise dilated convolution with a dilation rate of 2. The global context extractor performs global average pooling followed by a two-layer Linear-ReLU-Linear-Sigmoid, whose hidden dimension is set to $1/16$ of the input channels. The outputs from the local and

surrounding extractors are concatenated channel-wise, then passed through batch normalization and a PReLU activation. The output channel count remains equal to the input channel count of the block. The detailed principle of the CG Block is illustrated in the lower part of Fig. 5 and can be summarized as follows:

$$Y = f_{join}(f_{loc}(X) : f_{sur}(X) : f_{glo}(X)) \quad (2)$$

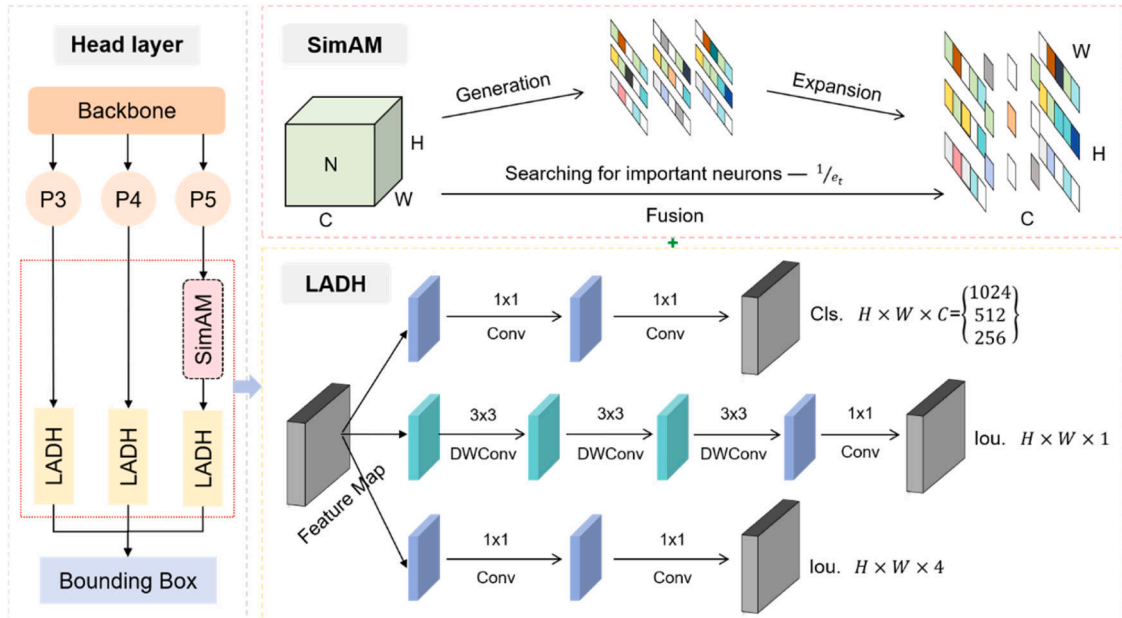


Fig. 6. A network architecture of the Lightweight Asymmetric Decoupled Head (LADH) with SimAM Attention Mechanism.

where, $f_{loc}(X)$ denotes the local feature extractor, which uses standard convolutional layers to learn fine-grained local features. $f_{sur}(X)$ denotes the surrounding context extractor, which utilizes dilated convolutional layers to capture context from a larger receptive field. $f_{glo}(X)$ denotes the global context extractor, which captures global information through global average pooling followed by a Multi-Layer Perceptron. $f_{join}(X)$ denotes the joint feature extractor, which refines the concatenated local and surrounding features using Batch Normalization and a Parametric ReLU activation.

3.2.2. Lightweight detection head network

As the core component that transforms feature maps into outputs for object detection tasks, the head detection network plays a crucial role in determining the model's accuracy, inference speed, and generalization capability. Building upon the asymmetric decoupled head (ADH) proposed by Zhang et al. [24], this study introduces a lightweight asymmetric decoupled head (LADH), as illustrated in Fig. 6. LADH employs three independent branches to handle different sub-tasks separately and integrates a multi-stage channel compression technique to reduce parameters within each convolutional layer, thereby significantly decreasing computational cost and model size.

To mitigate the interference of background noise, this study incorporates the lightweight attention mechanism SimAM [25] into the highest detection layer of the head network. The SimAM attention module introduces no extra parameters. It estimates the importance of each neuron in the feature map by optimizing an energy function through a linear transformation operation, enabling simultaneous attention across both channel and spatial dimensions. Through direct processing, the output dimension strictly remains the same as the input. This module is inserted after the highest-level feature map of the detection head to enhance the detection capability of large-object layers. First, the initial energy function for a single neuron is defined as:

$$e_t(w_t, b_t, y, x_i) = (y_t - \hat{t})^2 + \frac{1}{M-1} \sum_{i=1}^{M-1} (y_o - \hat{x}_i)^2 \quad (3)$$

Subsequently, e_t undergoes regularization, and its closed-form solution is derived to obtain a simplified expression e_t^* , which is used to identify key neuronal parameters, denoted as:

$$e_t^* = \frac{4(\hat{\sigma}^2 + \lambda)}{(t - \hat{\mu})^2 + 2\hat{\sigma}^2 + 2\lambda} \quad (4)$$

Finally, a scaling operator is applied to refine the feature map, enabling the model to focus more precisely on critical features. This effectively suppresses interference from background noise, thereby enhancing performance in the tomato detection task, represented as:

$$\tilde{X} = \text{sigmoid}\left(\frac{1}{E}\right) \odot X \quad (5)$$

As illustrated in Fig. 6, the SimAM attention mechanism is incorporated into the highest detection layer to enhance the representational capacity of high-level semantic features. This integration improves both detection accuracy and robustness, thereby increasing the model's adaptability in scenarios involving complex backgrounds and occlusions.

3.3. Multi-view compensation tracking and counting network

In complex scenarios, real-time multi-object tracking and counting often encounter multiple interferences such as occlusion, motion dynamics, and viewpoint variations. To overcome the limitations of traditional approaches, this study introduces the Deep OC-SORT algorithm and proposes a multi-view compensation (MVC) method. By integrating multiple collaborative modules, the robustness and accuracy of both tracking and counting are significantly enhanced.

3.3.1. Deep OC-SORT algorithm

Deep OC-SORT is a multi-object tracking algorithm that integrates deep learning with Kalman filtering. The algorithm comprises three key modules: Camera Motion Compensation (CMC), Dynamic Appearance (DA), and Adaptive Weighting (AW). These modules are cascaded to enable real-time tracking in dynamic scenes. The algorithm utilizes a pre-trained convolutional neural network to extract target appearance descriptors and fuses appearance information with motion cues through weighted similarity measures. During data association, the Hungarian algorithm is employed to solve the optimal assignment problem, while Kalman filtering enhances robustness in tracking fast-moving objects.

After the object detector produces a set of bounding boxes from a single frame, the DA module refines the detection results by extracting highly discriminative Re-Identification (ReID) features using a deep neural network, and constructing target appearance representation vectors. Existing trajectories are predicted using Kalman filtering, after which the CMC module applies an affine transformation matrix to geometrically correct the predicted states, thereby mitigating state drift caused by camera motion and enabling more accurate motion estimation. Finally, the AW module dynamically adjusts the fusion weights between motion information (Mahalanobis distance) and appearance information (cosine distance) based on the uncertainty represented by the covariance matrix output from the Kalman filter. This results in an optimized association cost matrix, thereby improving data association accuracy in complex scenarios. The overall pipeline is illustrated in Fig. 7.

3.3.2. Multi-view compensation method

To address the issues of false detections and missed detections during inspection robot operations, this study proposes an algorithmic framework based on multi-view compensation (MVC). By performing real-time multi-view cooperative detection and tracking on the video streams captured by the robot, the method significantly improves counting accuracy within continuous spatial regions. As illustrated in Fig. 8, the morphological features of the same target in two-dimensional images vary markedly due to differences in spatial position. Therefore, the detection area in this study is divided into three subregions located at different positions. The MVC method partitions each frame into distinct regions and performs multi-view cooperative detection and compensation, thereby decomposing the complex global multi-object tracking task into a set of parallel local subtasks.

The implementation of the MVC method proceeds as follows: first, a spatial partitioning module generates K adjacent subregions; second, each frame is processed by the cascaded network to produce unified outputs for ID assignment; finally, in the cross-view compensation module, contextual information across subregions is utilized to refine counting decisions. The method is built upon two key principles: (1) Local Appearance Consistency - by exploiting the fact that target

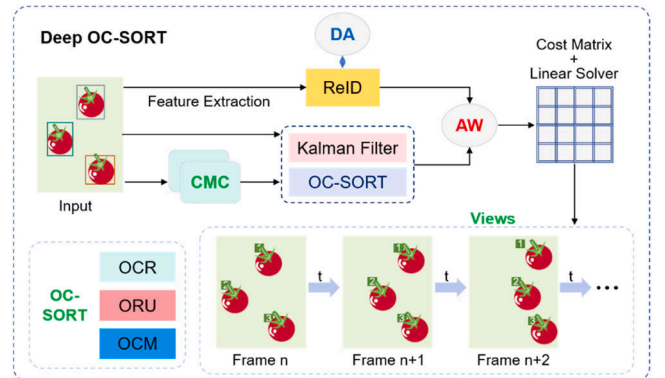


Fig. 7. The pipeline of Deep OC-SORT for tomato detection, tracking and counting.

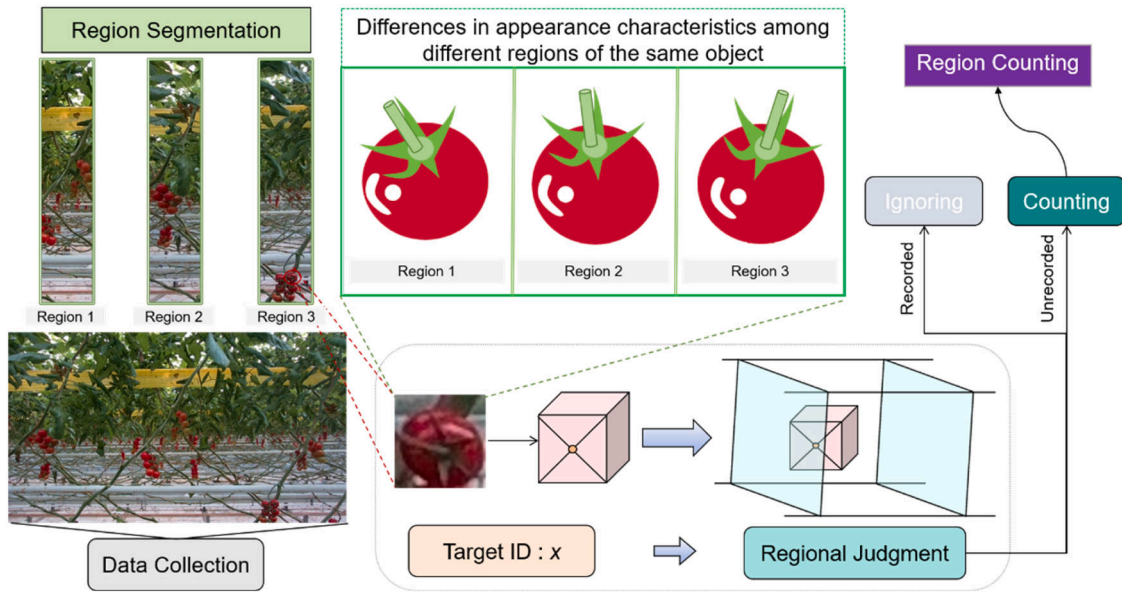


Fig. 8. Principle of the Multi-View Compensation (MVC) method based on spatial partitioning and redundant feature complementarity.

projection deformations remain nearly invariant within a limited spatial neighborhood, the stringent requirement for long-term tracking stability is relaxed to a weaker constraint on local tracking stability, thereby effectively suppressing identity-switch errors caused by long-distance target movement; and (2) Multi-View Redundant Complementarity — by performing independent detection within each subregion, targets occluded in one view can be effectively detected in other complementary views, and cooperative counting across the three regions substantially reduces missed detections while enhancing algorithmic robustness. Within each subregion, the system detects target IDs, and whether an ID changes is determined by a comprehensive cost matrix. This matrix fuses motion information (Mahalanobis distance) and appearance feature similarity (cosine distance), and the optimal assignment is solved using the Hungarian algorithm. To ensure tracking quality, a detection confidence threshold is set to 0.5: only detections with confidence higher than 0.5 are fed into the tracker; low-confidence detections are directly filtered out, reducing false alarms. Additionally, an IoU matching threshold of 0.3 is defined: when the Intersection over Union (IoU) of two bounding boxes exceeds 0.3, the algorithm considers them as potentially the same object and allows subsequent identity matching. If the IoU is equal to or below 0.3, they are treated as different targets and no association is performed. Under short-distance tracking conditions, where the effects of occlusion and illumination variation are relatively small, the target ID remains stable without switching, thereby enabling stable tracking over short ranges.

The Deep OC-SORT algorithm exhibits clear limitations when handling multi-target scenarios with complex backgrounds and highly variable target postures. The integration of the MVC method effectively addresses these deficiencies by employing a multi-region parallel detection and counting strategy. This not only mitigates false and missed detections caused by local occlusions or restricted viewpoints but also improves adaptability and accuracy in dense fruit distributions and complex greenhouse scenes with severe background interference, enabling a more comprehensive and precise assessment of tomato production conditions in greenhouses.

3.4. Edge device deployment

The greenhouse inspection robot deployed is equipped with the Jetson Orin Nano as its main control unit—an edge computing device developed by NVIDIA that combines low power consumption with high-

performance AI inference and computational capability. To ensure efficient operation under resource-constrained hardware conditions, this study applies lightweight architecture optimization and parameter tuning to both the backbone and detection head of the model, thereby achieving high detection performance while maintaining computational efficiency. Furthermore, TensorRT is employed to accelerate model inference. Techniques such as weight and activation precision calibration, layer and tensor fusion, and multi-stream execution are utilized to significantly enhance inference performance without compromising accuracy. The model is first converted from ONNX format to a TensorRT engine using the trtexec command-line tool provided by TensorRT, and then deployed for inference using TensorRT Runtimes. The deployment workflow of the developed multi-task cascaded neural network on the edge device is illustrated in Fig. 9.

4. Result and analysis

4.1. Model training and evaluation

4.1.1. Experiment environment

This study adopts a cloud-based training and edge-side testing strategy, in which model training is conducted on a cloud server, while testing and deployment are performed on a local PC. To ensure consistency across experimental conditions, the hardware configurations are closely matched. The input image size for the model is set to 640×640 pixels. Considering the influence of model depth and width on the number of parameters and computational complexity, the batch size is

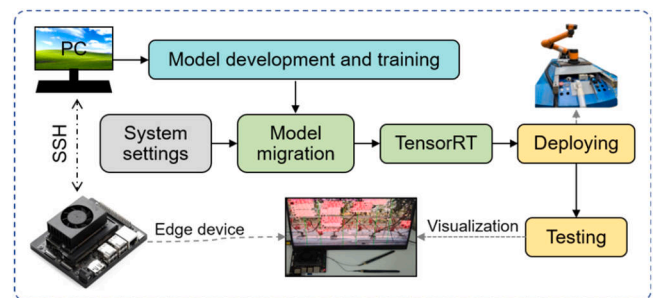


Fig. 9. Illustration of the deployment process and practical performance demonstration on the edge device Jetson Orin Nano.

set to 64, with an initial learning rate of 0.00672. The momentum and weight decay coefficients are configured to 0.86304 and 0.00241, respectively. To reduce the risk of overfitting, an early stopping mechanism is employed: training is automatically terminated if the mean Average Precision (mAP) does not exhibit a significant improvement over 50 consecutive epochs. The experimental dataset is based on a self-constructed cherry tomato dataset, and a transfer learning approach is adopted to enhance both training efficiency and model performance.

4.1.2. Evaluation indicators

To comprehensively evaluate the performance of the proposed model, this study adopts a range of evaluation indicators covering both object detection and object tracking and counting.

(1) Object Detection Indicators. For detection performance, the following indicators are selected: Precision, Recall, F1 Score, mAP@0.5, Model Size, Number of Parameters and FLOPs.

Precision (P) measures the proportion of correctly predicted positive samples among all positive predictions, defined as:

$$P = \frac{TP}{TP + FP} \quad (6)$$

where TP (True Positive) denotes correctly identified positive samples, and FP (False Positive) denotes negative samples incorrectly predicted as positive.

Recall (R) assesses the model's ability to detect all actual positive samples:

$$R = \frac{TP}{TP + FN} \quad (7)$$

where FN (False Negative) refers to positive samples incorrectly predicted as negative.

Mean Average Precision (mAP) is used as the primary indicator of object detection performance. It reflects the average precision across all classes at an IoU threshold of 0.5, providing a direct comparison of detection effectiveness:

$$AP = \int_0^1 P(R) dr \quad (8)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP \quad (9)$$

F1 Score offers a balanced assessment of model performance by harmonizing Precision and Recall:

$$F1 = \frac{2 \times P \times R}{P + R} \quad (10)$$

FLOPs represent the computational cost and complexity of the model, typically measured in GFLOPs. Number of Parameters and Model Size are critical indicators of model complexity, memory footprint, and suitability for real-time deployment on edge devices.

(2) Object Tracking and Counting Indicators. For tracking and counting performance, three commonly used indicators are employed: MOTA, HOTA, and IDF1.

Multiple Object Tracking Accuracy (MOTA) combines false negatives, false positives, and identity switches to evaluate overall tracking performance:

$$MOTA = 1 - \frac{|FN| + |FP| + |IDSW|}{|GT|} \quad (11)$$

where FN is the number of missed detections, FP the number of false positives, $IDSW$ the number of identity switches, and GT the total

number of ground truth targets.

Higher Order Tracking Accuracy (HOTA) provides a balanced assessment of both detection and association accuracy. It is calculated as:

$$HOTA = \sqrt{\frac{\sum_{c \in \{TP\}} A(c)}{|TP| + |FN| + |FP|}} \quad (12)$$

where $A(c)$ represents the accuracy at each association threshold c .

The ID F1 Score (IDF1) measures the accuracy of identity assignment by incorporating both ID precision and recall:

$$IDF1 = \frac{2 \cdot IDTP}{2 \cdot IDTP + IDFP + IDFN} \quad (13)$$

where $IDTP$ is the number of correctly identified trajectories, $IDFP$ denotes ID mismatches, and $IDFN$ refers to unmatched ground truth tracks.

Mean Absolute Error (MAE) quantifies the average of the absolute differences between predicted and true values across all individual samples:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (14)$$

where N is the total number of samples. y_i is the Ground Truth (GT) value for the i th sample $\hat{y}_i$ is the predicted value for the i th sample. $|y_i - \hat{y}_i|$ the Absolute Error (AE) for the i th sample.

Root Mean Square Error (RMSE) measures the square root of the average of the squared differences between predicted and true values

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (15)$$

where $(y_i - \hat{y}_i)^2$ is the Squared Error (SE) for the i th sample.

Weighted Mean Absolute Error (WMAE) is a variant of MAE that assigns different weights or levels of importance to the error of each sample when calculating the average absolute error

$$WMAE = \frac{\sum_{i=1}^N w_i \times |y_i - \hat{y}_i|}{\sum_{i=1}^N w_i} \quad (16)$$

where w_i is the weight assigned to the i th sample. $w_i \times |y_i - \hat{y}_i|$ is the Weighted Absolute Error for the i th sample. $\sum_{i=1}^N w_i$ is the sum of all weights.

Counting Precision is defined as $1 - \frac{|PT-GT|}{GT}$, expressed as a percentage. It measures the relative accuracy of the final count independent of identity association.

4.2. Result analysis of model training

During robotic inspection, performance is often influenced by factors such as occlusion, illumination variability, and other environmental disturbances. The mean Average Precision (mAP) serves as a comprehensive metric to evaluate the model's effectiveness under such complex conditions. In real-world operation scenarios, challenges such as background clutter and dense fruit overlap impose a dual requirement on the detection model: accurately identifying all fruits ("no misses") while avoiding false identification of non-fruit objects ("no false positives"). A low Recall would lead to underestimation of crop yield, whereas a low Precision could result in yield overestimation. The F1 Score, as a harmonic mean of Precision and Recall, provides an effective measure of how well the model balances these two objectives. Consequently, this study places particular emphasis on analyzing both mAP and F1 Score to comprehensively evaluate model performance.

As shown in Fig. 10, a comparative analysis between Model 6 and the baseline Model 0 reveals that the F1 Score curve of Model 6 is slightly lower during the early stages of training. However, as training progresses, Model 6 demonstrates improved stability in the mid-training phase, validating its robustness under various complex conditions. Notably, Model 6 achieves a higher peak performance during the later training stages, suggesting further optimization potential. With respect to mAP, although Model 0 shows strong early-stage performance, it exhibits overfitting in the mid-training phase. By contrast, while Model 6 initially underperforms slightly, it eventually matches and surpasses the peak mAP of Model 0 in subsequent training epochs. This behavior highlights the superior generalization capability and training stability of Model 6. Overall, across multiple training rounds, Model 6 consistently exhibits stronger comprehensive performance and better adaptability to complex visual environments.

To achieve accurate fruit localization, the metric Distance Focal Loss (DFL) is specifically emphasized in this study. DFL quantifies the localization precision of detected objects, and is particularly relevant to the accuracy of fruit counting and tracking. Accordingly, this study analyzes and compares the values of train/df_loss and val/df_loss, corresponding to the training and validation sets, respectively, to validate the methodological reliability. The results indicate that by epoch 200, both Model 0 and Model 6 show highly similar df_loss values, suggesting that the adopted detection strategy does not introduce additional localization error during fruit counting. This indirectly confirms the effectiveness and stability of the proposed method in accurately localizing tomato positions.

4.3. Ablation and comparison experiments

To validate the effectiveness of the proposed optimization strategies, a series of experiments were conducted using Precision, Recall, mAP@50, mAP@50–95, Model Size, Number of Parameters, and GFLOPs as evaluation indicators. The results of the ablation experiment are presented in Table 1.

Experimental results indicate that Model 1, by incorporating a lightweight module, achieved a significant reduction in both number of parameters and computational cost, while simultaneously improving the mAP@50 score. This result suggests that the lightweight design provides targeted improvements for object detection in occluded environments. Model 2, which employs a novel backbone replacement strategy, not only enhanced detection precision substantially but also achieved significant reductions in model parameters and computation. In addition, it maintained high performance across all key evaluation metrics. After integrating the LDAH module, Model 4 demonstrated improved Recall performance during the lightweight transformation process, further validating the effectiveness of this module design. A comparison between Model 5 and Model 6 reveals that the introduction of an attention mechanism can provide performance gains without increasing the number of parameters or computational overhead, highlighting its value in efficiency-sensitive scenarios.

Compared to Model 0 through Model 4, Model 6 stands out in terms of parameter efficiency, with a 38.9% reduction in the number of parameters compared to the baseline Model 0 (from 3.16 million to 1.93 million). Correspondingly, the model size decreased from 6.24 MB to

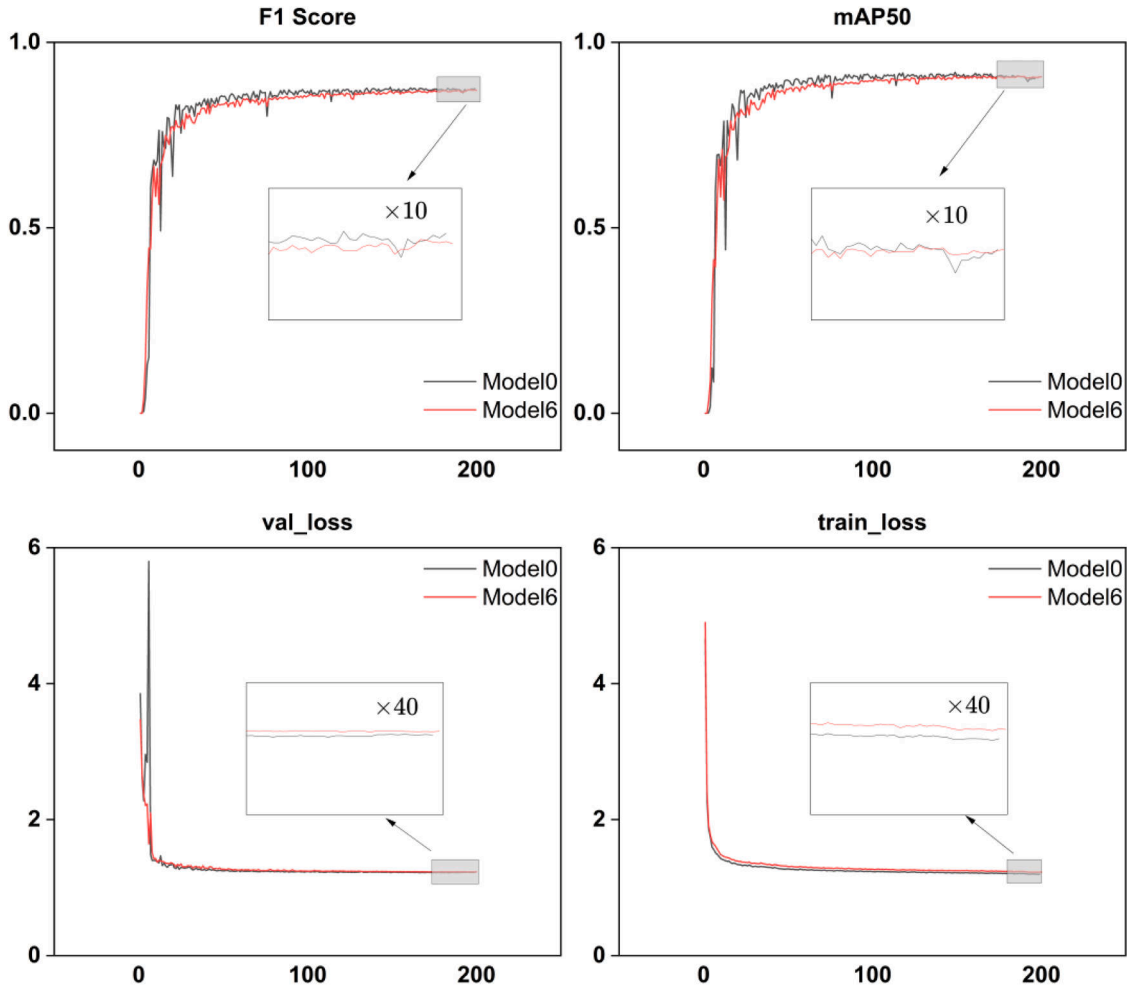


Fig. 10. The training results of the baseline model and the proposed model on F1_Score, mAP50, val_loss, train loss.

Table 1
Results of ablation experiment.

Models	CG	G-FConv	SimAM	LADH	P/%	Recall/%	mAP50/%	mAP50 -95/%	Size/MB	Params/10 ⁶	GFlops
Model0 (Baseline)	—	—	—	—	88.3	86.6	90.5	61.3	6.24	3.157200	8.9
Model1	✓	—	—	—	88.0	86.7	91.2	61.0	5.37	2.691773	7.5
Model2	—	✓	—	—	88.7	85.8	91.3	61.5	4.08	1.996440	6.7
Model3	—	—	✓	—	88.0	86.5	90.9	61.2	6.24	3.157200	8.9
Model4	—	—	—	✓	87.8	87.2	91.3	61.4	6.21	3.139968	8.1
Model5	✓	✓	—	✓	86.8	85.8	90.3	59.1	3.98	1.939637	5.3
Model6 (Our)	✓	✓	✓	✓	88.3	86.2	90.8	59.8	3.98	1.939637	5.3

3.98 MB, a 36.2% reduction, and the GFLOPs dropped by 40.4%, making it the most computationally efficient among all models tested. This lightweight architecture not only improves training efficiency but also greatly reduces the demands for storage and computational resources, thereby enhancing its deployability on edge devices. In terms of performance, Model 6 outperforms Model 0 in mAP while consuming fewer resources during both inference and training, making it especially suitable for real-time applications with limited hardware capacity. Additionally, the introduction of the SimAM module in Model 6 led to superior performance over Model 5 in terms of Precision, Recall, and mAP@50, indicating that SimAM significantly contributes to noise suppression and enhancement of key feature representation. Notably, despite its reduced computational complexity and model size, Model 6 maintains a level of Precision comparable to the baseline Model 0. Although there is a slight decrease in Recall, the values of mAP@50 and mAP@50-95 remain relatively stable. This confirms the reliability of Model 6 in retaining detection accuracy during the lightweight transformation.

As illustrated in Fig. 11, Model 6 achieves substantial improvements in lightweight metrics such as model size, computational cost, and number of parameters. More importantly, it attains a slight improvement in the key metric mAP for tomato yield estimation while maintaining its lightweight nature, thereby effectively balancing model compactness and detection performance optimization.

To further validate the superiority of our model, we compared it with several representative detectors, including SSD, YOLOv8n, MobileNet-YOLO, and YOLOv9t, under identical dataset and training settings. As summarized in Table 2, the proposed model achieves a competitive mAP@0.5 of 90.8%, slightly lower than YOLOv9t (91.2%) but on par with YOLOv8n (90.5%), and substantially higher than MobileNet-YOLO (88.8%) and SSD (56.3%). More importantly, our model achieves the smallest model size (3.98 MB), the lowest computational cost (5.3 GFLOPs), and the fewest parameters (1.94 M) among all compared methods. Compared to YOLOv8n, our model reduces model size by 36%, FLOPs by 40%, and parameters by 38%. These results demonstrate that the proposed G-FConv + L-MFA + LADH architecture strikes an excellent balance between detection accuracy and lightweight efficiency, making it particularly suitable for real-time deployment on edge inspection robots.

Table 2
Performance comparison with different detectors.

Model	Precision	mAP@0.5	Parameters (Million)	FLOPs (G)	size (MB)
SSD	83.8	56.3	23.880	30.5	99.1
YOLOv8n	88.3	90.5	3.1572	8.9	6.24
MobileNet-YOLO	87.2	88.8	2.5004	6.4	5.04
YOLOv9t	88.4	91.2	2.1287	8.5	4.43
Proposed	88.3	90.8	1.9396	5.3	3.98

4.4. Visualization analysis of detection results

To further investigate the model's decision-making mechanism and enhance its interpretability, this study employed Gradient-weighted Class Activation Mapping (Grad-CAM) [26] to visualize the regions of interest during the tomato detection task. Grad-CAM generates class-discriminative heatmaps that highlight the spatial regions contributing most significantly to the model's predictions, thereby providing intuitive visual evidence for evaluating the rationality of decision-making and diagnosing potential errors.

The visualization results are presented in Fig. 12. Overall, both the proposed improved model and the baseline model are capable of focusing their attention on tomato target regions; however, the improved model demonstrates a clear performance advantage. Under normal illumination conditions, the activation regions of the improved model are more concentrated and primarily centered on the core areas of tomato clusters, whereas the baseline model exhibits relatively dispersed and blurred activations. In low-light scenarios, the improved model maintains strong and well-defined responses, while the baseline model shows weakened activations and incomplete coverage. Under strong illumination conditions, the improved model accurately distinguishes and localizes multiple independent tomato clusters, whereas the baseline model is more susceptible to background interference. In summary, the improved model exhibits superior attention concentration capability and robustness across diverse environmental conditions. Although some sensitivity persists under extremely complex scenarios, its overall performance remains within the acceptable operational range.

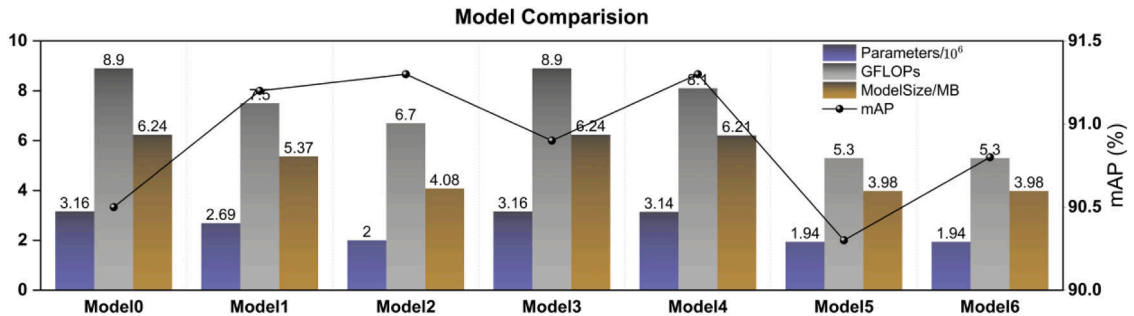


Fig. 11. Comparison of key indicators for different models.

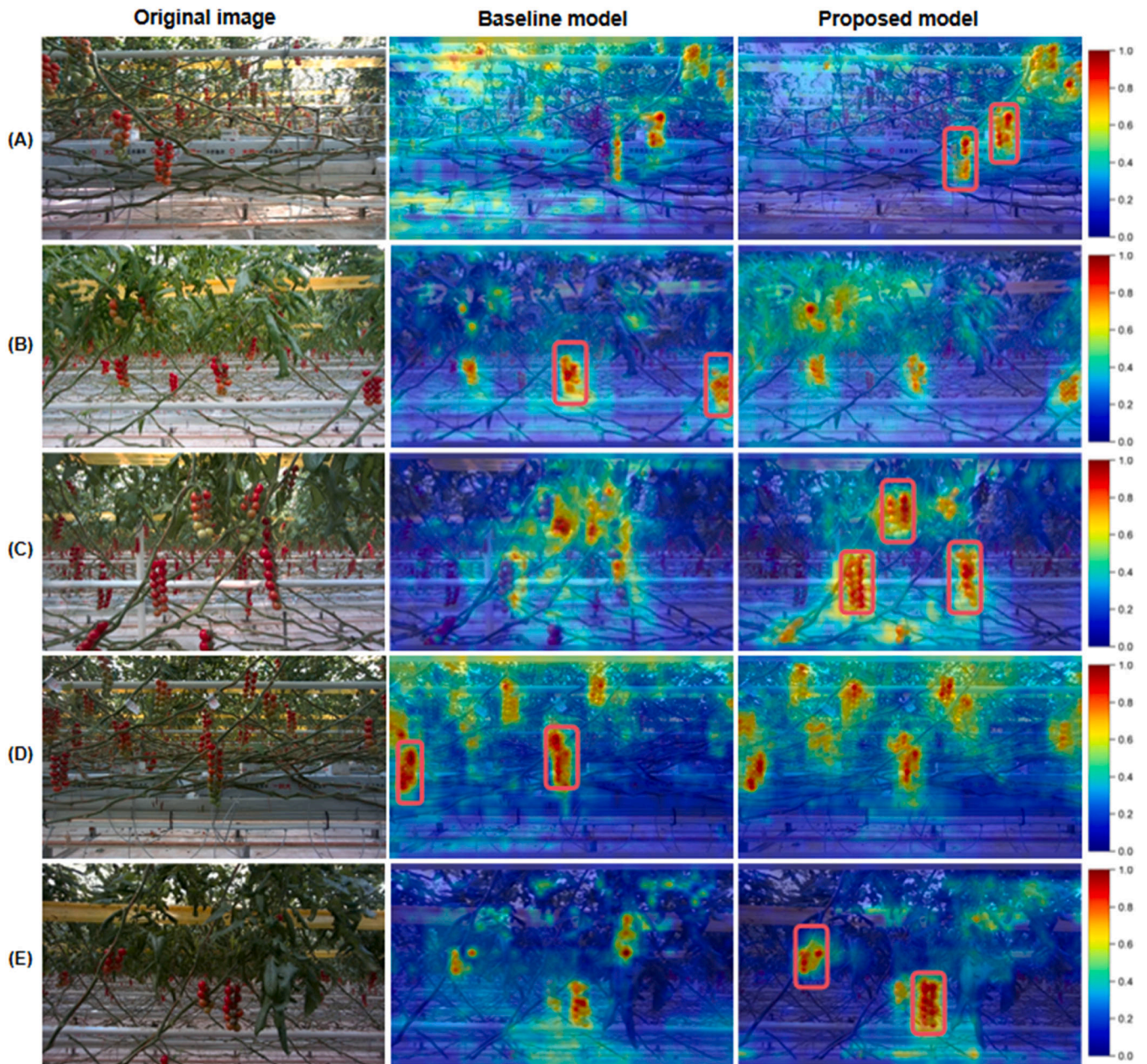


Fig. 12. Grad-CAM Visualization of Model Focus Across Diverse Scenarios. This figure presents a comparative analysis of visual heatmaps generated by Grad-CAM for the Baseline and Proposed models, illustrating feature attention under varied environmental conditions.

for greenhouse inspection tasks.

4.5. Comparative analysis of tracking and counting results

In this study, a custom dataset designed for tomato tracking tests was annotated with ID labels using the DarkLabel tool. The model's tracking performance was evaluated using four metrics: MOTA, HOTA, IDF1, and DetA (detection accuracy). The results are summarized Table 3. In crowded greenhouse scenes, conventional multi-object tracking metrics

Table 3

Multi-object tracking performance evaluation.

Model	MOTA/%	HOTA/%	IDF1/%	DetA /%
Baseline	−8.97	8.36	10.47	68.2
Our	−8.77	8.56	11.05	71.5

such as MOTA, HOTA, and IDF1 are strongly penalized by frequent occlusions and reappearances, often yielding low or even negative values. For example, a temporarily occluded fruit that is later re-detected is counted as both a false negative (during occlusion) and a false positive (upon reappearance), causing rapid error accumulation that does not necessarily reflect counting performance. Therefore, we place greater emphasis on DetA, which measures detection quality independently of identity association. As shown in Table 3, the proposed model achieves a DetA of 71.5%, outperforming the baseline (68.2%). This 3.3-point improvement confirms that our lightweight detection architecture (G-FConv + L-MFA + LADH + SimAM) enhances fruit localization robustness under challenging greenhouse conditions. Although the HOTA and IDF1 scores remain relatively low (8.56% and 11.05%, respectively) due to severe occlusion, dense fruit clustering, and camera motion, the improved DetA demonstrates that the primary bottleneck lies in association, not detection – a key motivation for our multi-view

compensation (MVC) strategy.

To validate the effectiveness of the proposed multi-region compensation strategy, this study evaluated SVC, MVC, and Deep OC-SORT across six distinct scenarios. The experimental results are presented in Table 4. The findings indicate that both compensation-based methods (SVC and MVC) significantly outperform the baseline tracking method, Deep OC-SORT, in counting performance. Specifically, the baseline method Deep OC-SORT exhibits substantial counting deviations, with a mean absolute error (MAE) as high as 595.03 and a root mean square error (RMSE) of 644.44. These results suggest that the baseline approach struggles to handle ID switching and duplicate counting in dense and heavily occluded environments. In sharp contrast, the adoption of compensation strategies in SVC and MVC leads to remarkable performance improvements, reducing the error values to approximately one-twentieth of those observed in Deep OC-SORT and achieving results closely aligned with the ground truth (GT). Furthermore, compared with the single-view compensation method (SVC), the multi-view compensation method (MVC) achieves lower MAE (26.11) and RMSE (30.50), corresponding to reductions of 15% and 10%, respectively. These results indicate that the MVC framework, through its multi-region compensation mechanism, effectively mitigates errors that remain uncorrected in SVC due to the inherent limitations of a single viewpoint. Even in severely occluded and highly dense scenarios (e.g., Seg 5 and Seg 6), the proposed method maintains relatively low error levels. The multi-view compensation mechanism effectively addresses tracking failures caused by occlusion and appearance similarity, thereby substantially enhancing the final counting accuracy.

To investigate the impact of frame rate on tracking performance, this study conducted a comparative analysis of the counting performance of the SVC and MVC methods under three frame rate settings: 10 FPS, 20 FPS, and 30 FPS. The detailed results are presented in Table 5. The experimental data indicate that as the frame rate increases from 10 FPS to 30 FPS, the mean absolute error (MAE) of SVC and MVC decreases by 25.8% and 43.7%, respectively. These findings demonstrate that tracking accuracy improves with higher frame rates. This improvement occurs because a higher frame rate reduces inter-frame displacement of targets, thereby lowering the probability of ID switches and missed detections, which in turn enhances counting stability. This improvement can be attributed to the reduced inter-frame motion variation at higher frame rates, which enables the model to obtain more stable positional information, thereby effectively reducing false negatives (FN) and enhancing counting robustness. Furthermore, MVC consistently outperforms SVC across all frame rate settings. In complex scenarios under low frame rate conditions (10 FPS), MVC achieves a 15.3% reduction in MAE and a 20.0% reduction in root mean square error (RMSE) compared with SVC. Under high frame rate conditions (30 FPS), MVC attains its best performance (MAE = 38.33; RMSE = 38.41). Relative to SVC, MAE and RMSE are reduced by 35.8% and 39.5%, respectively, further widening the performance gap. These results confirm that the MVC method effectively compensates for missed detections in single-view settings through multi-view feature fusion, demonstrating strong robustness and environmental adaptability.

In complex greenhouse environments, illumination variation constitutes one of the critical factors affecting crop counting performance. To evaluate the adaptability of the proposed model under different greenhouse production scenarios, two distinct video sequences were selected, and comparative experiments were conducted under five artificially controlled illumination conditions (50%, 75%, 100%, 125%, and 150%). As illustrated in the radar chart in Fig. 13, both counting methods exhibit moderate performance under low-light conditions, which may be attributed to the inherent accuracy limitations of the baseline detection model. However, under high-illumination conditions, and even in overexposed scenarios (e.g., brightness levels set to 125% and 150%), the MVC algorithm significantly outperforms SVC. These experimental results demonstrate that the MVC algorithm maintains superior robustness and adaptability under strong and dynamically varying illumination conditions.

To evaluate the counting performance across different target categories, this study conducted counting experiments on two custom-defined classes—ripe tomatoes (“rt”) and unripe tomatoes (“ut”)—across six distinct scenarios. The results are presented in Table 6. As shown in the table, the baseline model exhibits substantial deviations in counting performance for both categories, with a weighted mean absolute error (WMAE) as high as 598.08. In contrast, the proposed SVC and MVC methods produce counting results that are considerably closer to the ground truth for both maturity levels. Notably, the MVC algorithm demonstrates superior counting accuracy for the ripe category, achieving a WMAE of only 20.10. Overall, the MVC algorithm exhibits clear advantages in detection and counting performance across different fruit maturity stages. It is particularly well-suited for yield estimation tasks targeting mature fruits, where higher accuracy and robustness are required.

To evaluate the counting performance of the proposed model under practical production conditions, model predictions were conducted in a real greenhouse environment, and the predicted results were subjected to regression analysis against the ground truth (GT). The results are presented in Fig. 14. As shown in the figure, the data points are closely distributed around the diagonal line, indicating a strong linear correlation between the predicted and actual values ($R^2 = 0.99$). The model maintains high counting consistency across varying fruit densities and degrees of occlusion. Furthermore, the fitted linear regression equation is $Y = 0.926X - 6.1605$, where the slope (0.926) is close to the ideal value of 1. This further confirms the strong agreement between the predicted and ground truth values. In summary, the proposed model demonstrates high accuracy and reliability in tomato yield estimation tasks and satisfies the practical application requirements of real-world greenhouse production environments.

4.6. Comparison among multiple counting methods and error source analysis

In the analysis of MOT metrics, it was found that unstructured scenes, camera motion, position and pose, and other factors have a serious impact on target tracking performance. To provide an intuitive

Table 4
The results of the comprehensive video testing and counting of multiple actual scenarios.

Video	GT	SVC			MVC			Deep OC-SORT		
		Count	MI	FP	Count	MI	FP	Count	MI	FP
Seg 1	58	50	2	3.47%	50	2	3.47%	189	132	228.03%
Seg 2	105	96	3	2.85%	102	2	1.90%	559	453	430.38%
Seg 3	229	203	6	2.62%	214	4	1.75%	908	679	296.29%
Seg 4	288	251	7	2.43%	251	7	2.43%	1034	746	259.03%
Seg 5	343	305	8	2.33%	305	8	2.33%	1254	911	265.29%
Seg 6	417	387	10	2.40%	395	10	2.40%	1067	650	155.88%
MAE		30.72			26.11			595.03		
RMSE		34.04			30.50			644.44		

Note: FP (False Positive rate) = MI / GT, where MI denotes misidentification in this counting method.

Table 5
Comparative performance of compensation methods under varying frame rates.

Video	GT	SVC					MVC				
		Count	MI	FP	MAE	RMSE	Count	MI	FP	MAE	RMSE
10FPS											
Video 1	464	342	10	2.16%	80.33	88.3	380	11	2.89%	68	70.69
Video 2	392	374	33	8.42%			374	33	8.82%		
Video 3	436	405	27	6.19%			405	27	6.67%		
20FPS											
Video 1	464	382	7	1.51%	63.67	66.26	418	12	2.87%	51.33	51.96
Video 2	392	350	4	1.02%			355	3	0.85%		
Video 3	436	387	7	1.61%			387	7	1.81%		
30FPS											
Video 1	464	379	5	1.08%	59.67	63.47	431	6	1.39%	38.33	38.41
Video 2	392	355	4	1.02%			355	4	1.13%		
Video 3	436	395	7	1.61%			408	7	1.72%		

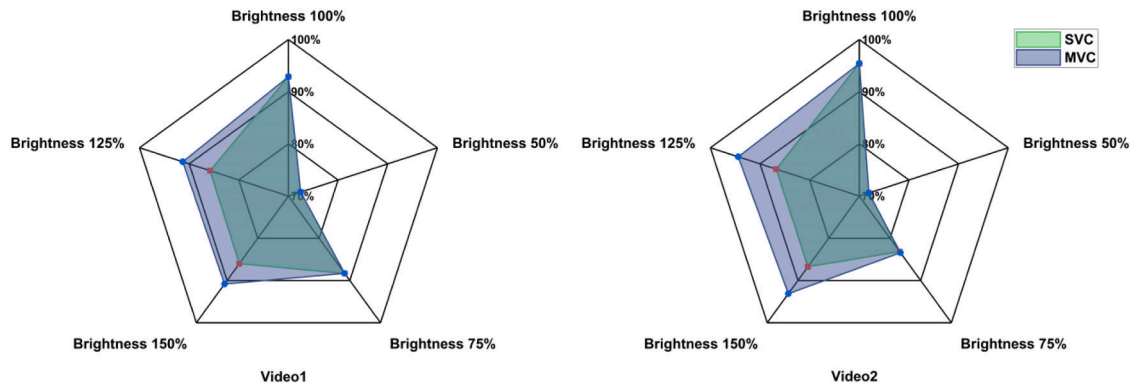


Fig. 13. Comparative robustness of SVC and MVC under varied illumination.

Table 6
Maturity-specific counting performance comparison.

Video	GT		SVC		MVC		Deep OC-SORT	
	ut	rt	ut	rt	ut	rt	ut	rt
Video1	118	206	93	180	93	184	588	708
Video2	104	231	82	212	82	214	652	1119
Video3	204	256	172	240	172	240	1011	903
Video4	172	217	143	201	143	204	800	837
Video5	137	262	105	246	108	248	560	1003
Video6	181	244	159	226	166	226	468	641
WMAE			21.88		20.10		598.08	

comparison of the counting performance among the three methods—ID-based counting, SVC, and MVC—five consecutive video frames were randomly selected for evaluation. The results are presented in Fig. 15. Under inter-frame analysis, yellow represents the original target ID, while red represents ID switches. It can be observed that ID switches occur from the first frame to the fifth frame due to various factors. Using the original ID-based counting approach, the majority of targets were successfully detected; however, instances of target loss occurred, leading to unstable counting results. In contrast, both the SVC and MVC methods achieved more accurate counting outcomes without missed counts, demonstrating greater stability throughout the detection and counting process. Compared with the SVC method, MVC incorporates a region-based compensation strategy, in which the counting area is divided into multiple independent subregions. Multi-view feature information is then complementarily fused to mitigate counting fluctuations caused by target tracking loss or ID switching under a single-view configuration. When a target is temporarily lost within a specific region due to occlusion or ID switching, it can be re-detected and counted in adjacent regions, or counting errors can be reduced through boundary-based logical

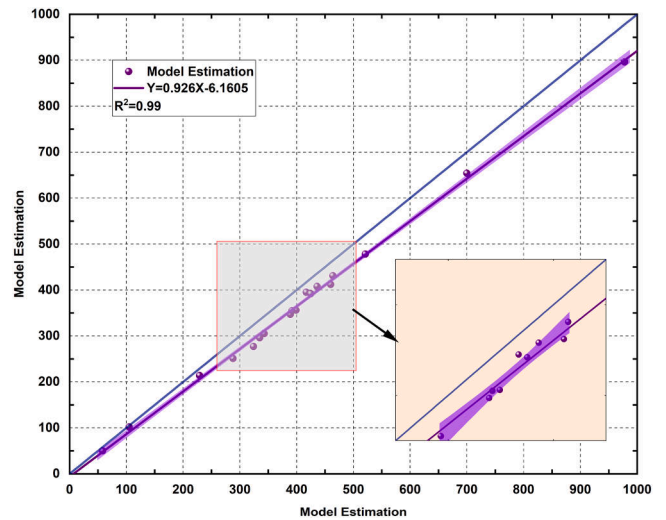


Fig. 14. Ground truth versus estimated number of tomatoes per plot. The horizontal axis refers to the number of tomatoes estimated by the proposed model, and the vertical axis refers to the number of tomatoes that have been manually counted.

constraints. By reducing sensitivity to ID loss, the MVC method significantly enhances reliability and precision in complex and dynamic greenhouse tomato production environments.

5. Discussion

In response to the demand for accurate prediction of tomato yield,

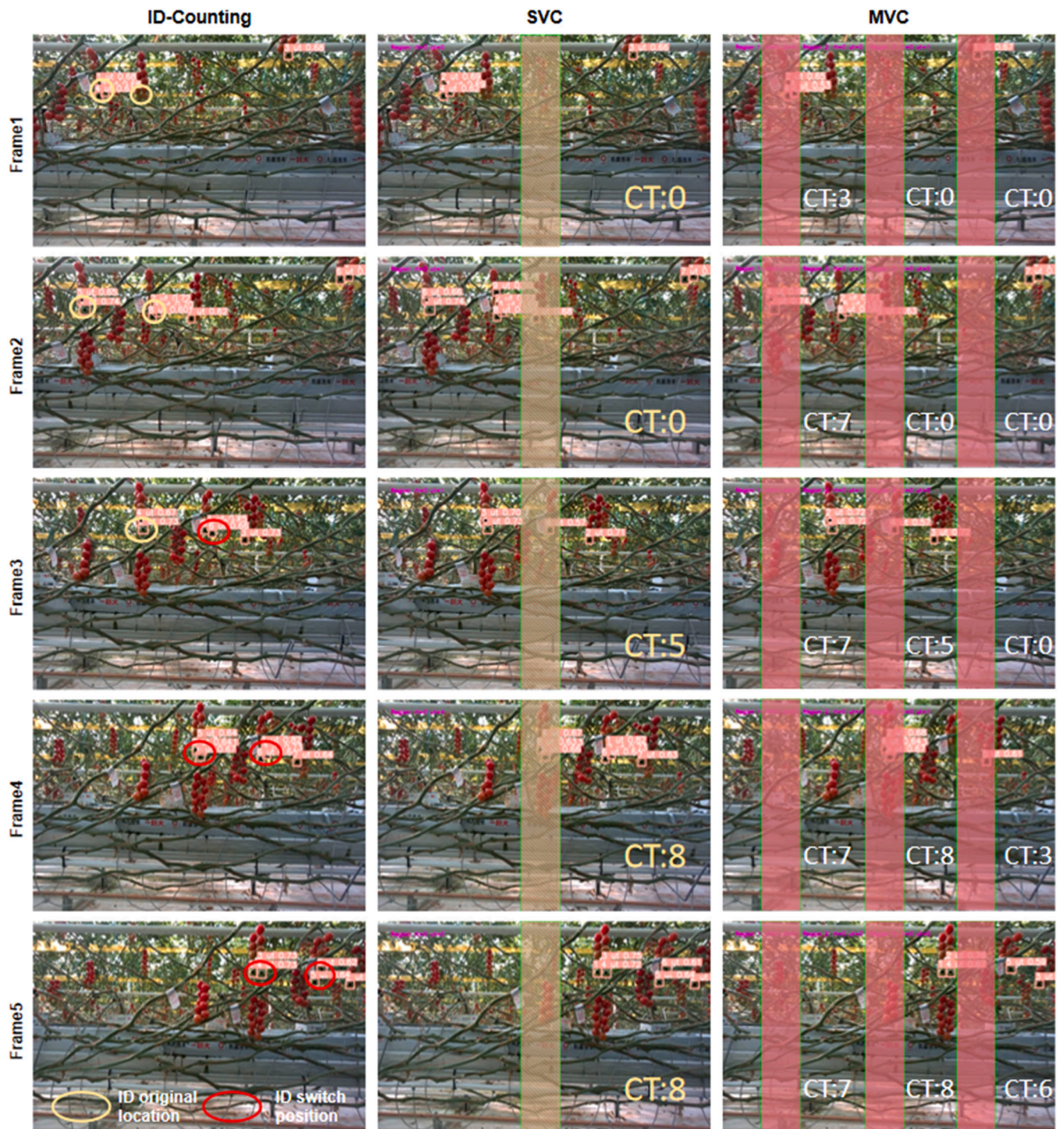


Fig. 15. Comparison of Counting Strategies Across Continuous Frames. This figure illustrates the counting results of five consecutive video frames, comparing the baseline ID-Counting, Single-View Compensation (SVC), and the proposed Multi-View Compensation (MVC) method.

this study proposes a lightweight multi-task cascaded neural network capable of real-time and precise detection, tracking, and counting of tomatoes in complex greenhouse environments. By improving the network architecture and optimizing parameters, the proposed method effectively addresses issues such as high computational cost and deployment on edge devices. A multi-view compensation strategy is introduced to enhance the accuracy of tracking and counting under challenging conditions such as occlusion, object overlap, and dynamic scenes. Comparative experiments conducted under diverse scenarios, illumination conditions, and frame rates (see Tables 4, 5, and Fig. 14)

validate the robustness of the proposed MVC method under complex operating conditions. Furthermore, multi-category counting analysis (see Table 6) demonstrates its significant advantage in classification-based counting tasks involving fruits at different maturity stages. In real production environments, the model's predicted results were compared with manual counting values, and linear regression analysis (see Fig. 15) further confirmed the effectiveness and reliability of the proposed method in practical application scenarios. In addition, instance-level analysis based on consecutive video frames indicates that, compared with SVC and Deep OC-SORT, the MVC method achieves

superior counting performance in complex greenhouse environments.

Compared with representative works in related domains, our method offers distinct advantages. For lightweight edge detection, models such as YOLOv8n and MobileNet-YOLO achieve high accuracy but either have larger model sizes or lack an integrated counting module. As shown in Table 2, our model achieves the smallest size (3.98 MB) and lowest computational cost (5.3 GFLOPs) while maintaining competitive mAP (90.8%). For multi-view crop phenotyping, many existing methods rely on 3D reconstruction from multiple calibrated cameras, which is computationally expensive and difficult to deploy on a moving robot. By contrast, our MVC works directly on 2D video streams from a single camera, using spatial partitioning and confidence-weighted fusion to reduce counting errors by 95.6% without 3D reconstruction. For agricultural robot perception, complete solutions often focus on detection and segmentation but do not specifically address the multi-view counting problem under severe occlusion. However, our framework uniquely combines a lightweight detection backbone, an occlusion-robust MVC counter, and a systematic error analysis, making it suitable for real-time yield estimation in greenhouse inspection robots.

In terms of real-time detection, the model significantly reduces the number of parameters while maintaining detection accuracy. This is achieved by integrating lightweight convolutional modules and attention mechanisms deeply into the backbone network, thereby effectively compressing computational complexity. Although the proposed method already achieves promising performance in detection, there remains room for improvement, such as expanding the dataset to include more complex environmental scenarios to enhance model robustness. Furthermore, the lightweight design involves a trade-off between model size and the extraction of fine-grained features, which may lead to reduced sensitivity in identifying extremely small or heavily blurred targets in the far field. Future work could explore more adaptive feature fusion techniques to maintain high precision across all scales without significant computational overhead.

For tracking and counting tasks, the study addresses the ID switch issue inherent in Deep OC-SORT by introducing the MVC method. Through a region-based divide-and-conquer strategy, this approach enables multi-angle feature complementation, substantially reducing counting errors in complex greenhouse scenes. Nevertheless, challenges such as ID switches caused by occlusion and overlapping, as well as positional state shifts of tomatoes due to camera motion across consecutive frames, still require further refinement and optimization. Additionally, the current multi-view compensation (MVC) strategy relies on fixed regional partitions, which might lack flexibility when the inspection robot undergoes non-uniform motion or drastic vibrations. The cumulative error in counting over extremely long planting rows also remains a technical bottleneck. Integrating inertial measurement units (IMU) data with visual odometry could potentially mitigate these motion-induced tracking failures in future iterations.

6. Conclusion

In response to the application requirements for multi-object real-time detection and accurate counting of tomatoes in complex greenhouse environments, this study proposes a lightweight cascaded framework integrating detection, tracking, counting, and compensation. Experimental results demonstrate that the proposed network achieves real-time processing performance of 31.3 FPS on an NVIDIA Jetson Orin Nano edge device, while maintaining high detection accuracy (90.8%) and counting precision (95.2%). In terms of model optimization, architectural refinements and parameter compression reduced computational cost to 5.3 GFLOPs (a 40.4% decrease) and shrank the model size to 3.98 MB, significantly enhancing the feasibility of edge deployment. The proposed MVC method utilizes a multi-view redundant complementarity mechanism to significantly suppress miscounting caused by identity switching. This approach effectively reduces the counting error in dense and occluded scenarios by 95.6%. Validation in

real-world production settings confirmed a yield prediction accuracy of 93%, providing robust technical support for dynamic harvest planning and supply chain loss reduction. The proposed method offers an innovative solution for tomato yield assessment in complex greenhouse environments and can be extended to the detection of other crop production scenarios, holding substantial theoretical significance and practical application potential.

Statement for studies in humans / animals

This study does not involve any research related to humans or animals.

CRediT authorship contribution statement

Xushuo Xing: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Jin Gao:** Writing – review & editing, Resources, Methodology, Funding acquisition. **Xue Deng:** Methodology. **Shubo Wang:** Investigation. **Peng Qi:** Resources, Investigation. **Fangyan Wang:** Resources, Methodology. **Xiuning Hou:** Writing – review & editing. **Junfeng Gao:** Writing – review & editing.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

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Data availability

The authors do not have permission to share data.

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